PROBLEM 21

Gravitation & Centrifugal Balance

Rotating Planet & Apparent Weight

Original statement · Problems.pdf, p. 10

On a spinning spherical planet of radius R and gravitational acceleration g_p, the bodies weight twice as much on the poles as on the equator. Determine the duration of the day on that planet.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A spherical planet of radius R spins with constant angular velocity $\boldsymbol{\Omega} = \Omega \mathbf{k}$ around its polar axis.
- **Gravitational Field:** Uniform surface gravitational acceleration g_p directed toward the center of the planet.
- **Apparent Weight:** The weight measured by a spring balance or scale is the normal reaction force $\mathbf{N}$ exerted by the surface to support a static mass m on the surface.
- **Problem Condition:** A body on the pole weighs twice as much as on the equator:

$$W_{pole} = 2W_{equator}$$

- **Objective:** Determine the duration of the planet's sidereal day $T_{day} = \frac{2\pi}{\Omega}$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame $\mathcal{F}_0$:** Frame centered at the planet center O.
- **Rotating Frame $\mathcal{F}_1$:** Planet-fixed reference frame rotating with angular velocity $\boldsymbol{\omega}_{1/0} = \Omega\mathbf{k}$.
- **Acceleration of a Stationary Point on the Surface:**

For a body resting on the surface, relative velocity and acceleration vanish: $\mathbf{v}' = \mathbf{0}, \mathbf{a}' = \mathbf{0}$.

The inertial acceleration is purely centripetal:

$$\mathbf{a}_0 = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = -\Omega^2 \boldsymbol{\rho}$$

where $\boldsymbol{\rho}$ is the projection of the position vector onto the equatorial plane.

- **Change of basis (planet-fixed rotating basis).** Let $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be the inertial basis and $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}\}$ the basis fixed to the planet, rotating at Ω about the polar axis $\mathbf{k}$, with $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} + \sin\Omega t\,\mathbf{j}$. The columns are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- A body fixed at the equator at longitude φ_0 has $\mathbf{r} = R\,[{}_0R_1](\cos\varphi_0, \sin\varphi_0, 0)^T = R\,\mathbf{e}_r$, with $\mathbf{e}_r = \cos(\Omega t + \varphi_0)\,\mathbf{i} + \sin(\Omega t + \varphi_0)\,\mathbf{j}$. Poisson's formula $\dot{\mathbf{e}}_r = \Omega\,\mathbf{e}_\varphi$, $\dot{\mathbf{e}}_\varphi = -\Omega\,\mathbf{e}_r$ gives $\mathbf{a}_0 = -\Omega^2 R\,\mathbf{e}_r$, the centripetal acceleration used in Phase 3. The determinant, RR^T and this acceleration were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

1. Dynamic Equilibrium in the Rotating Frame

Pedagogical Justification: In the non-inertial reference frame of the planet, static equilibrium requires the sum of real gravitational force $\mathbf{F}_g$, reaction force $\mathbf{N}$, and fictitious centrifugal force $\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = m\Omega^2\boldsymbol{\rho}$ to vanish.

$$\mathbf{F}_g + \mathbf{N} + \mathbf{F}_{cf} = \mathbf{0} \implies \mathbf{N} = -\mathbf{F}_g - \mathbf{F}_{cf}$$

The apparent weight is the magnitude of the normal force, $W = |\mathbf{N}|$.

2. Apparent Weight at the Pole

At the geographic poles, the position vector lies along the rotation axis $\mathbf{k}$:

$$\mathbf{r}_{pole} = \pm R\mathbf{k} \implies \boldsymbol{\rho}_{pole} = \mathbf{0}$$

The centrifugal force vanishes at the poles:

$$\mathbf{F}_{cf,pole} = \mathbf{0}$$

Therefore:

$$\mathbf{N}_{pole} - mg_p\mathbf{k} = \mathbf{0} \implies W_{pole} = mg_p$$

3. Apparent Weight at the Equator

At the equator, the position vector lies entirely in the equatorial plane:

$$|\boldsymbol{\rho}_{eq}| = R$$

Gravity acts radially inwards: $\mathbf{F}_{g,eq} = -mg_p\mathbf{e}_r$.

The centrifugal force acts radially outwards: $\mathbf{F}_{cf,eq} = +m\Omega^2 R\mathbf{e}_r$.

Normal reaction from the surface: $\mathbf{N}_{eq} = N_{eq}\mathbf{e}_r$.

Equilibrium equation:

$$-mg_p\mathbf{e}_r + N_{eq}\mathbf{e}_r + m\Omega^2 R\mathbf{e}_r = \mathbf{0}$$

$$N_{eq} = m(g_p - \Omega^2 R)$$

Thus, the apparent weight at the equator is:

$$W_{equator} = m(g_p - \Omega^2 R)$$

4. Computation of the Rotation Rate Ω

Imposing the given condition $W_{pole} = 2W_{equator}$:

$$mg_p = 2m(g_p - \Omega^2 R)$$

Divide both sides by m:

$$g_p = 2g_p - 2\Omega^2 R$$

Rearrange to isolate Ω^2:

$$2\Omega^2 R = 2g_p - g_p = g_p$$

$$\Omega^2 = \frac{g_p}{2R} \implies \Omega = \sqrt{\frac{g_p}{2R}}$$

5. Duration of the Day

The duration of one full planetary rotation (the day) is:

$$T_{day} = \frac{2\pi}{\Omega} = \frac{2\pi}{\sqrt{\frac{g_p}{2R}}} = 2\pi\sqrt{\frac{2R}{g_p}} = 2\sqrt{2}\pi\sqrt{\frac{R}{g_p}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Analysis:**

$[T_{day}] = \sqrt{\frac{[R]}{[g_p]}} = \sqrt{\frac{\mathrm{m}}{\mathrm{m/s^2}}} = \sqrt{\mathrm{s^2}} = \mathrm{s}.$

The result is dimensionally sound.

- **Centrifugal Breakup Limit:**

A planet becomes gravitationally unbounded at the equator (weightlessness) when $\Omega_{crit} = \sqrt{g_p/R}$. Here $\Omega = \sqrt{g_p/(2R)} = \frac{1}{\sqrt{2}}\Omega_{crit} \approx 0.707\,\Omega_{crit}$, ensuring that objects remain resting on the equatorial surface.

- **Numerical illustration (Python):** for $g_p = 9.81\ \mathrm{m/s^2}$ and $R = 6.371 \times 10^6\ \mathrm{m}$, $\Omega = \sqrt{g_p/(2R)} = 8.774 \times 10^{-4}\ \mathrm{rad/s}$ and $T_{day} = 2\pi\sqrt{2R/g_p} = 7.161 \times 10^3\ \mathrm{s} = 1.99\ \mathrm{h}$, which satisfies $\Omega^2 R/g_p = 0.5$ (so $W_{equator} = \frac{1}{2}W_{pole}$ as imposed).

Source: Problems.pdf (Page 10)